Real QZ Is Sufficient for DSGE Determinacy

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Abstract

Some DSGE solvers reject parameter values when complex-output QZ produces imaginary entries in the policy matrix. I show that this is not a determinacy condition. QZ gives a basis for the initial values of stable paths. Let Z_x be the part of this basis associated with predetermined variables. A unique stable solution exists if and only if the number of stable roots equals the number of predetermined variables and Z_x is nonsingular. Real and complex QZ span the same stable subspace, so they give the same test and the same real solution in exact arithmetic. Complex roots occur in conjugate pairs and describe real damped cycles. In one million benchmark draws, an imaginary-part restriction rejects 464,685 draws accepted by real QZ.

Keywords: DSGE models; determinacy; generalized Schur decomposition; QZ decomposition; GPU computing.

1 Introduction

The QZ representation should not change which parameter values admit a unique stable solution. A numerical rule in the repository nevertheless makes the classification depend on that representation. In the Smets–Wouters benchmark, real QZ accepts 702,789 of one million prior draws. A rule based on complex-output QZ accepts only 238,104. Both routines count the same stable roots. The difference comes from a rule that rejects an imaginary entry larger than 10^{-9} in the recovered solution.

Result. Let n_s be the number of stable roots. Let Z_x contain the state rows of a basis for the stable initial values. I show that every predetermined initial state has exactly one stable path if and only if n_s equals the number of predetermined variables and Z_x is nonsingular. Real and complex QZ produce the same stable subspace. They therefore give the same root count, the same rank condition, and the same real first-order solution in exact arithmetic. An imaginary-part restriction is not part of the determinacy test.

Intuition. The predetermined state x_0 is inherited from the past. The jump variables y_0 adjust so that the economy follows a stable path. QZ collects the stable initial values in a subspace. If c_0 gives the coordinates of a point in that subspace, its state component must satisfy

$$x_0 = Z_x c_0.$$

The model has a unique stable solution precisely when this equation has one solution c_0 for every x_0. With one state and one jump variable, the stable initial values form a line in the (x, y) plane.

A nonvertical line can be written as $y = Gx$, so each x has one associated y. A vertical line does not have this property. In many dimensions, the stable subspace must likewise be the graph of a function from states to jumps. The root count matches the dimensions; nonsingularity of Z_x makes the mapping one-to-one and onto.

Complex roots. Complex roots do not change this geometry. The model matrices are real, so non-real roots occur in conjugate pairs. A pair describes a real cycle. Real QZ stores it in a real 2×2 block; complex QZ stores it in two complex columns. These are two bases for the same stable subspace. In exact arithmetic, both bases give the same real policy rule.

Numerical issue. In complex arithmetic, the imaginary parts from a conjugate pair must cancel. Roundoff can make the cancellation incomplete, especially when Z_x is ill-conditioned. A remaining imaginary part is therefore a numerical diagnostic, not an economic restriction.

The result follows from the Blanchard–Kahn dimension condition and the graph construction used in generalized-Schur solution methods [1, 3, 5]. Section 2 derives the determinacy condition. Section 3 explains complex roots and the numerical problem. Section 4 reports the benchmark evidence. The appendix gives the formal proof.

2 The determinacy condition

2.1 States, jumps, and stable paths

Fix the model parameters and a path for the fundamental shocks. To study uniqueness, subtract any two candidate solutions. Their difference solves the homogeneous system

$$Dw_{t+1} = Ew_t, \qquad w_t = \begin{bmatrix} x_t \\ y_t \end{bmatrix}. \tag{1}$$

The vector $x_t \in \mathbb{R}^{n_x}$ contains the predetermined variables. Its initial value x_0 is fixed by the past. The vector y_t contains jump variables. Its initial value can adjust to place the economy on a stable path. A stable path satisfies equation (1) and converges to zero. Throughout, uniqueness is conditional on the fundamental shocks and excludes additional sunspot shocks.

Assumption 1 (Regularity)**.** *The generalized eigenvalue problem is regular, no finite root lies on the unit circle, and admissible initial values satisfy any algebraic restrictions implied by the model.*

Regularity means that $\det(E - \lambda D)$ is not identically zero. The unit-circle restriction separates stable roots from unstable roots. The last condition matters when D is singular. In that case the system may contain static restrictions, represented by infinite generalized roots. These roots are not stable dynamic roots.

To see where the roots come from, consider a path of the form $w_t = \lambda^t v$. Substitution into equation (1) gives

$$(E - \lambda D)v = 0.$$

Thus λ is a transition root of the pencil $E - \lambda D$. The path converges when $|\lambda| < 1$.

The QZ decomposition sorts the system's dynamic directions by their roots. Real QZ writes

$$Q^{\mathsf{T}} DZ = S, \qquad Q^{\mathsf{T}} EZ = T, \tag{2}$$

where Q and Z are real orthogonal matrices. The repository records the transition roots as $\lambda_i = \beta_i/\alpha_i$. After QZ places the stable roots first, the first n_s columns of Z form a basis Z_s for all

stable initial values. This set is the stable subspace. Appendix A proves this statement, including the case of a singular D.

2.2 One equation determines existence and uniqueness

Partition the stable basis into rows for states and jumps:

$$Z_s = \begin{bmatrix} Z_x \\ Z_y \end{bmatrix}. \tag{3}$$

Every stable initial value can be written as

$$\begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} Z_x \\ Z_y \end{bmatrix} c_0.$$

The vector $c_0 \in \mathbb{R}^{n_s}$ gives the coordinates of the initial value within the stable subspace. Since x_0 is predetermined, the model must solve

$$x_0 = Z_x c_0. \tag{4}$$

Equation (4) is the source of the determinacy condition.

Proposition 1 (Existence and uniqueness)**.** *Under Assumption 1, every initial state $x_0 \in \mathbb{R}^{n_x}$ has exactly one stable path if and only if*

$$n_s = n_x \qquad \textit{and} \qquad \operatorname{rank}(Z_x) = n_x.$$

Equivalently, Z_x is square and nonsingular.

Why the condition is sufficient. Suppose $n_s = n_x$ and Z_x has full rank. Then Z_x is invertible. For any initial state, equation (4) has the unique solution $c_0 = Z_x^{-1} x_0$. This coordinate vector selects one point in the stable subspace and therefore one stable path.

Why the condition is necessary. Now suppose every initial state has exactly one stable path. Existence for every x_0 means that the map $c_0 \mapsto Z_x c_0$ is onto: it can reach every state. Uniqueness means that the map is one-to-one: two coordinate vectors cannot produce the same state. A linear map is both onto and one-to-one only when its domain and codomain have the same dimension and its matrix has full rank. Hence $n_s = n_x$ and Z_x is nonsingular.

The failures also have simple interpretations. If $n_s < n_x$, there are too few stable directions, so some initial states have no stable path. If $n_s > n_x$, there are too many stable directions, so a feasible state has more than one stable path. If $n_s = n_x$ but Z_x is singular, some states are infeasible and feasible states have multiple stable paths.

When the condition holds, the jump variables are a function of the states. Substituting $c_t = Z_x^{-1} x_t$ into $y_t = Z_y c_t$ gives the policy rule

$$y_t = G x_t, \qquad G = Z_y Z_x^{-1}. \tag{5}$$

The stable subspace is therefore the graph of G. The appendix also derives the state-transition matrix.

3 Why complex roots are not a problem

3.1 A complex pair is a real cycle

Because D and E are real, $\det(E - \lambda D)$ has real coefficients. A non-real root λ must therefore appear with its conjugate $\bar{\lambda}$. The same is true of the associated generalized vectors.

A two-variable example gives the intuition. Let

$$M = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}, \qquad a + ib = \rho e^{i\phi}.$$

The eigenvalues of M are $a \pm ib$, but M is real. Its powers are

$$M^t = \rho^t \begin{bmatrix} \cos(t\phi) & -\sin(t\phi) \\ \sin(t\phi) & \cos(t\phi) \end{bmatrix}. \tag{6}$$

The angle ϕ determines the cycle, and ρ determines its size. If $\rho < 1$, the cycle shrinks over time. The roots are complex, but the path is real and stable.

3.2 Real and complex QZ use different coordinates

Real and complex QZ span the same stable subspace. They only use different bases. Let Z_s be the real basis and let

$$\widetilde{Z}_s = Z_s C$$

be a complex basis for the same subspace, where C is nonsingular. The state and jump blocks are then $\widetilde{Z}_x = Z_x C$ and $\widetilde{Z}_y = Z_y C$.

If the solution is unique, the policy rule computed from the complex basis is

$$\widetilde{Z}_y \widetilde{Z}_x^{-1} = Z_y C (Z_x C)^{-1} = Z_y Z_x^{-1} = G. \tag{7}$$

The change of basis cancels. Since the same policy rule can be computed from a real basis, the exact policy rule is real. Appendix C shows the same result for the state-transition matrix.

Corollary 1 (Real and complex QZ)**.** *Under Assumption 1, real and complex ordered QZ give the same stable-root count and the same rank condition in exact arithmetic. When the solution is unique, they give the same real first-order solution.*

3.3 Where the numerical problem enters

Complex-output QZ represents a conjugate pair with two complex columns. In exact arithmetic, their imaginary contributions cancel in equation (7). In floating-point arithmetic, the two columns need not be exact conjugates. Solving with an ill-conditioned Z_x can magnify the resulting error.

This explains why a complex calculation can leave an imaginary part in a policy matrix that is real in theory. The imaginary part measures numerical error in the chosen representation. Rejecting the parameter draw on this basis changes the determinacy test. Real QZ avoids the cancellation because it keeps each conjugate pair in a real 2×2 block [4].

4 Evidence from the HLT benchmark

The audit draws one million parameter vectors from the independent bounded package priors for the 18 parameters in `phase1_18params`. The model is `medium_sw07_hlt`. The script `benchmarks/qz_cayley_large_support_audit.py` produces `benchmarks/results/qz_real_cayley_hlt_loose_1m_stream.json`.

Real QZ is the reference calculation. It checks the stable-root count, the rank of Z_x, and the relative residual of the quadratic matrix equation

$$AX^2 + BX + C = 0.$$

The tolerance for the residual is 10^{-8}. The old complex-output rule adds a bound of 10^{-9} on the largest imaginary entry in X.

Table 1: Determinacy classifications for one million benchmark prior draws

Event	Count	Share
Real QZ: unique stable solution	702,789	70.3%
Real QZ: indeterminate	256,850	25.7%
Real QZ: no stable solution	40,361	4.0%
Complex-output QZ: unique stable solution	238,104	23.8%
Real QZ accepts, complex-output rule rejects	464,685	46.5%
Complex-output rule accepts, real QZ rejects	0	0.0%
Cayley real-part rule disagrees with real QZ	0	0.0%
Doubling accepts, real QZ rejects	84,606	8.5%

Notes: Shares use all 1,000,000 draws as the denominator. Among draws accepted by real QZ, the median relative residual is 8.82×10^{-14} and the maximum is 1.04×10^{-11}. The acceptance tolerance is 10^{-8}. The rows after the first three are comparisons and do not form a partition of the sample.

Table 1 shows that the two QZ routines agree on the stable-root count. All 464,685 disagreements come from the imaginary-part restriction. Real QZ produces a real solution for these draws, with residuals below the acceptance tolerance.

The Cayley real-part rule agrees with real QZ on all one million draws. This is evidence for the tested model and prior region, not a proof for every pencil. Doubling solves the matrix equation for every draw accepted by real QZ, but it also accepts 84,606 draws that fail the stable-root count. A small equation residual verifies a candidate matrix root. It does not prove that the stable solution is unique.

4.1 Implication for GPU computation

The determinacy result requires a stable subspace, not a particular QZ routine. This distinction gives a real GPU algorithm that does not compute eigenvectors. Define

$$H = (D - E)^{-1}(D + E).$$

If λ is a model root, the corresponding root of H is $\mu = (1 + \lambda)/(1 - \lambda)$. Moreover,

$$\mathrm{Re}(\mu) = \frac{1 - |\lambda|^2}{|1 - \lambda|^2}.$$

Thus $|\lambda| < 1$ if and only if $\mathrm{Re}(\mu) > 0$.

The matrix sign maps the roots of H in the right half-plane to $+1$ and the roots in the left half-plane to -1. Therefore

$$P_s = \frac{I + \operatorname{sign}(H)}{2}$$

is the projector onto the stable subspace. Its dimension is $(m + \operatorname{tr}(\operatorname{sign}(H)))/2$, where m is the order of the pencil. This gives the stable-root count and a basis for the stable subspace using real arithmetic only.

The implementation computes the matrix sign by the scaled Newton iteration

$$X_{k+1} = \frac{1}{2}\left(\gamma_k X_k + \gamma_k^{-1} X_k^{-1}\right), \qquad X_0 = H,$$

where γ_k balances the norms of X_k and X_k^{-1}. This standard iteration converges to the matrix sign when H has no root on the imaginary axis [2]. The code uses real matrix products and linear solves, then checks the sign residual, the stable-root count, and the QME residual.

The matrix-sign method does not compute a generalized QZ decomposition. It computes only the stable subspace needed for the first-order solution. CPU real QZ remains the reference for numerical audits, especially near the unit circle or when the state block of the stable subspace is poorly conditioned.

5 Conclusion

The determinacy test is simple. The stable subspace must contain exactly one point for each predetermined state. This holds if and only if the number of stable roots equals the number of predetermined variables and Z_x is nonsingular.

Complex roots do not add another condition. They describe real cycles and can be represented by either real or complex coordinates. Real QZ is therefore sufficient for the determinacy test and the first-order solution.

A Why the stable subspace contains all stable paths

Order the QZ decomposition so that the stable finite blocks appear first:

$$S = \begin{bmatrix} S_s & S_{su} \\ 0 & S_u \end{bmatrix}, \qquad T = \begin{bmatrix} T_s & T_{su} \\ 0 & T_u \end{bmatrix}, \qquad Z = \begin{bmatrix} Z_s & Z_u \end{bmatrix}.$$

Write $w_t = Zc_t$ and partition $c_t = (c_{s,t}', c_{u,t}')'$. The lower block of $Sc_{t+1} = Tc_t$ evolves independently of $c_{s,t}$. Its finite transition roots have modulus greater than one. A path that converges to zero must therefore have zero coordinates on these blocks. Assumption 1 removes the coordinates associated with infinite roots through the model's algebraic restrictions.

The stable block satisfies

$$S_s c_{s,t+1} = T_s c_{s,t}.$$

The roots in this block are finite, so S_s is nonsingular. Define $A_s = S_s^{-1} T_s$. Every eigenvalue of A_s has modulus below one, which implies $A_s^t \to 0$. Hence every $Z_s c_{s,0}$ generates a stable path, and every stable path starts in range(Z_s).

B Proof of Proposition 1

Let P_x project an initial value onto its predetermined variables. On the stable subspace,

$$P_x(Z_s c_0) = Z_x c_0.$$

Thus a stable path exists for x_0 if and only if $Z_x c_0 = x_0$ has a solution. Existence for every x_0 is equivalent to Z_x being onto.

Suppose $Z_x h = 0$ for some $h \neq 0$. If c_0 gives a stable path for x_0, then $c_0 + h$ gives a second stable path with the same x_0. The two initial values differ because Z_s has full column rank. Thus uniqueness requires Z_x to be one-to-one.

Conversely, if Z_x is one-to-one, two stable paths with the same x_0 must have the same c_0. Their later values are also equal because $c_{s,t+1} = A_s c_{s,t}$. Hence existence and uniqueness for every x_0 are equivalent to Z_x being both onto and one-to-one.

The domain of Z_x has dimension n_s, and its codomain has dimension n_x. The map is onto and one-to-one if and only if $n_s = n_x$ and $\operatorname{rank}(Z_x) = n_x$. This proves the proposition. □

C Proof of Corollary 1

The generalized eigenvalues are roots of the pencil, so they do not depend on a QZ basis. Since D and E are real, non-real roots occur in conjugate pairs. The stable set is closed under conjugation. Real and complex ordered QZ therefore select the same stable subspace and the same number n_s of stable roots.

Let Z_s and $\widetilde{Z}_s$ be real and complex bases for this subspace. There is a nonsingular matrix C such that $\widetilde{Z}_s = Z_s C$. It follows that

$$\operatorname{rank}(\widetilde{Z}_x) = \operatorname{rank}(Z_x C) = \operatorname{rank}(Z_x).$$

The two decompositions give the same determinacy test.

When the solution is unique, equation (7) shows that both bases give the same policy matrix. The state transition is also basis-invariant. In the real stable coordinates it is

$$L = Z_x A_s Z_x^{-1}.$$

In the complex coordinates, $\widetilde{A}_s = C^{-1} A_s C$. Therefore

$$\widetilde{Z}_x \widetilde{A}_s \widetilde{Z}_x^{-1} = Z_x C (C^{-1} A_s C)(Z_x C)^{-1} = Z_x A_s Z_x^{-1} = L.$$

The real basis produces real G and L. The complex basis produces the same matrices in exact arithmetic. This proves the corollary. □

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